Lesson 15: Metric Spaces

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

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1 Why Functional Analysis?

Reinforcement learning algorithms operate on *function spaces*. A value function $V\colon \mathcal{S} \to \mathbb{R}$ assigns a real number to every state; a Q-function $Q\colon \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ assigns a value to every state-action pair. When we write the Bellman optimality equation

$$V^*(s) = \max_{a \in \mathcal{A}} \left[r(s,a) + \gamma \sum_{s'} p(s' \mid s,a)\, V^*(s') \right], \tag{1}$$

we are asserting that V^* is a *fixed point* of an operator T acting on a space of functions. The fundamental questions are:

1. Does a fixed point exist?
2. Is it unique?
3. Does the iteration $V_{n+1} = TV_n$ converge to V^*?
4. How fast does it converge?

Answering these questions requires the language of functional analysis:

- **Metric spaces** (Section 2) provide the notion of distance needed to say "V_n converges to V^*."
- **Normed spaces** (Section **??**) give us a notion of "size" for functions and a way to measure error $\|V_n - V^*\|$.
- **Completeness** (Section **??**) ensures that Cauchy sequences converge, which is essential for the Banach fixed-point theorem.
- **Bounded linear operators** (Section **??**) formalize the maps between function spaces that arise in approximation theory.
- **Compactness** (Section **??**) is the key to existence results and understanding the behavior of infinite-dimensional spaces.
- **The Baire category theorem** (Section **??**) underpins the three pillars of linear functional analysis: the Uniform Boundedness Principle, the Open Mapping Theorem, and the Closed Graph Theorem.

The Bellman Operator as Motivation

The Bellman optimality operator T defined by the right-hand side of (1) acts on the Banach space $(b(\mathcal{S}), \|\cdot\|_\infty)$ of bounded real-valued functions on the state space, equipped with the supremum norm. The contraction mapping theorem (Banach fixed-point theorem, which we prove in the next lesson) guarantees that T has a unique fixed point and that $\|T^n V_0 - V^*\|_\infty \le \gamma^n \|V_0 - V^*\|_\infty$. Understanding *why* this works requires every concept in this lesson: metric, norm, completeness, and the operator framework.

2 Metric Spaces

2.1 Definition and Examples

Definition 2.1 (Metric Space)**.** A *metric* on a set X is a function $d\colon X \times X \to [0,\infty)$ satisfying, for all $x, y, z \in X$:

(M1) $d(x,y) = 0$ if and only if $x = y$. (identity of indiscernibles)
(M2) $d(x,y) = d(y,x)$. (symmetry)
(M3) $d(x,z) \le d(x,y) + d(y,z)$. (triangle inequality)

The pair (X,d) is called a *metric space*. Elements of X are called *points*.

Remark 2.1. Non-negativity $d(x,y) \ge 0$ follows from the axioms: by (M3) and (M1), $0 = d(x,x) \le d(x,y) + d(y,x) = 2d(x,y)$ by (M2), so $d(x,y) \ge 0$.

Example 2.1 (Euclidean Metric)**.** $X = \mathbb{R}^n$ with $d(x, y) = \left(\sum_{i=1}^n (x_i - y_i)^2\right)^{1/2}$. The triangle inequality follows from the Cauchy–Schwarz inequality.

Example 2.2 (p-Metrics on $\mathbb{R}^n$)**.** For $1 \le p \le \infty$, define $d_p(x, y) = \|x - y\|_p$ where

$$\|x\|_p = \begin{cases} \left(\sum_{i=1}^n |x_i|^p\right)^{1/p} & \text{if } 1 \le p < \infty, \\ \max_{1 \le i \le n} |x_i| & \text{if } p = \infty. \end{cases}$$

Each d_p is a metric. The triangle inequality for $1 < p < \infty$ is Minkowski's inequality (proved in Proposition **??**).

Example 2.3 (Discrete Metric)**.** On any set X, define $d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ if $x \neq y$. This is a metric. Every subset of X is open (and closed), so X with the discrete metric is a *discrete topological space*.

Example 2.4 (Function Space $C[a, b]$). $]Let$C[a,b]$denotethesetofcontinuousreal-valuedfunctionson$[a,b]$.Def$ $\max_{t \in [a,b]} |f(t) - g(t)|$, $\quad d_1(f, g) = \int_a^b |f(t) - g(t)|\, dt$. Both are metrics on $C[a, b]$. The metric d_∞ measures uniform distance; d_1 measures average distance.

Example 2.5 (Sequence Spaces ℓ^p)**.** For $1 \le p < \infty$, define $\ell^p = \{(x_n)_{n=1}^\infty \subset \mathbb{R} : \sum_{n=1}^\infty |x_n|^p < \infty\}$ with metric $d(x, y) = \left(\sum_{n=1}^\infty |x_n - y_n|^p\right)^{1/p}$. For $p = \infty$, $\ell^\infty = \{(x_n) : \sup_n |x_n| < \infty\}$ with $d(x, y) = \sup_n |x_n - y_n|$.

State Spaces as Metric Spaces

In RL with a finite state space $\mathcal{S} = \{s_1, \ldots, s_n\}$, a value function $V \colon \mathcal{S} \to \mathbb{R}$ is simply a vector in $\mathbb{R}^n$. The space of all value functions is $(\mathbb{R}^n, \|\cdot\|_\infty)$, which is a metric space. For infinite or continuous state spaces, value functions live in function spaces like $C(\mathcal{S})$ or $\ell^\infty(\mathcal{S})$, and the theory of this section becomes essential.

2.2 Topology of Metric Spaces

Definition 2.2 (Open Ball)**.** In a metric space (X, d), the *open ball* of radius $r > 0$ centered at $x \in X$ is

$$B(x, r) = \{y \in X : d(x, y) < r\}.$$

The *closed ball* is $\overline{B}(x, r) = \{y \in X : d(x, y) \le r\}$.

Definition 2.3 (Open and Closed Sets)**.** Let (X, d) be a metric space.

(i) A set $G \subseteq X$ is *open* if for every $x \in G$ there exists $r > 0$ such that $B(x, r) \subseteq G$.
(ii) A set $F \subseteq X$ is *closed* if its complement $X \setminus F$ is open.

Proposition 2.1 (Properties of Open and Closed Sets)**.** *In a metric space (X, d):*

(a) $\emptyset$ and X are both open and closed.
(b) Any union of open sets is open.
(c) Any finite intersection of open sets is open.
(d) Any intersection of closed sets is closed.
(e) Any finite union of closed sets is closed.
(f) Every open ball $B(x, r)$ is open.

Proof. **(a)** $\emptyset$ is open vacuously. X is open because $B(x, r) \subseteq X$ for all x and $r > 0$.

(b) Let $\{G_\alpha\}_{\alpha \in I}$ be a family of open sets and let $x \in \bigcup_\alpha G_\alpha$. Then $x \in G_\beta$ for some β, so there exists $r > 0$ with $B(x, r) \subseteq G_\beta \subseteq \bigcup_\alpha G_\alpha$.

(c) Let $G_1, \ldots, G_n$ be open and let $x \in \bigcap_{i=1}^n G_i$. For each i, there exists $r_i > 0$ with $B(x, r_i) \subseteq G_i$. Set $r = \min(r_1, \ldots, r_n) > 0$. Then $B(x, r) \subseteq \bigcap_{i=1}^n G_i$.

(d)–(e) Follow from (b)–(c) by taking complements (De Morgan's laws).

(f) Let $y \in B(x,r)$ and set $\varepsilon = r - d(x,y) > 0$. For any $z \in B(y,\varepsilon)$, we have $d(x,z) \leq d(x,y) + d(y,z) < d(x,y) + \varepsilon = r$, so $z \in B(x,r)$. Thus $B(y,\varepsilon) \subseteq B(x,r)$. □

Definition 2.4 (Interior, Closure, Boundary)**.** Let $A \subseteq X$ in a metric space (X,d).

(i) The *interior* of A is $\operatorname{int}(A) = \bigcup\{G \subseteq A : G \text{ is open}\}$.
(ii) The *closure* of A is $\operatorname{cl}(A) = \bigcap\{F \supseteq A : F \text{ is closed}\}$.
(iii) The *boundary* of A is $\partial A = \operatorname{cl}(A) \setminus \operatorname{int}(A)$.
(iv) A is *dense* in X if $\operatorname{cl}(A) = X$.

Proposition 2.2 (Closure Characterization)**.** *$x \in \operatorname{cl}(A)$ if and only if for every $r > 0$, $B(x,r) \cap A \neq \emptyset$. Equivalently, $x \in \operatorname{cl}(A)$ if and only if there exists a sequence (x_n) in A with $x_n \to x$.*

Proof. ($\Rightarrow$) Suppose $x \in \operatorname{cl}(A)$ and there exists $r > 0$ with $B(x,r) \cap A = \emptyset$. Then $A \subseteq X \setminus B(x,r)$, which is closed. So $\operatorname{cl}(A) \subseteq X \setminus B(x,r)$, contradicting $x \in \operatorname{cl}(A)$.

($\Leftarrow$) Suppose every ball around x meets A, and let F be any closed set containing A. If $x \notin F$, then $x \in X \setminus F$, which is open, so there exists $r > 0$ with $B(x,r) \subseteq X \setminus F$. But then $B(x,r) \cap A \subseteq B(x,r) \cap F = \emptyset$, contradicting our assumption. So $x \in F$ for every closed $F \supseteq A$, hence $x \in \operatorname{cl}(A)$.

For the sequential characterization: if $B(x,1/n) \cap A \neq \emptyset$ for each n, pick $x_n \in B(x,1/n) \cap A$; then $d(x_n,x) < 1/n \to 0$. Conversely, if $x_n \to x$ with each $x_n \in A$, then every ball $B(x,r)$ contains x_n for large n, so $B(x,r) \cap A \neq \emptyset$. □

2.3 Convergence and Completeness

Definition 2.5 (Convergence)**.** A sequence (x_n) in a metric space (X,d) *converges* to $x \in X$ if $d(x_n,x) \to 0$ as $n \to \infty$, i.e., for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $d(x_n,x) < \varepsilon$ for all $n \geq N$. We write $x_n \to x$ or $\lim_{n\to\infty} x_n = x$.

Proposition 2.3 (Uniqueness of Limits)**.** *In a metric space, limits are unique: if $x_n \to x$ and $x_n \to y$, then $x = y$.*

Proof. $0 \leq d(x,y) \leq d(x,x_n) + d(x_n,y) \to 0 + 0 = 0$, so $d(x,y) = 0$, hence $x = y$. □

Definition 2.6 (Cauchy Sequence)**.** A sequence (x_n) in (X,d) is a *Cauchy sequence* if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $d(x_m,x_n) < \varepsilon$ for all $m,n \geq N$.

Proposition 2.4. *Every convergent sequence is Cauchy.*

Proof. If $x_n \to x$, then for $\varepsilon > 0$ there exists N such that $d(x_n,x) < \varepsilon/2$ for $n \geq N$. For $m,n \geq N$: $d(x_m,x_n) \leq d(x_m,x) + d(x,x_n) < \varepsilon/2 + \varepsilon/2 = \varepsilon$. □

Cauchy Does Not Imply Convergent

The converse fails in general. In $\mathbb{Q}$ with the usual metric, the sequence of rational approximations to $\sqrt{2}$ (e.g., defined by $x_1 = 1$, $x_{n+1} = (x_n + 2/x_n)/2$) is Cauchy but does not converge in $\mathbb{Q}$. The "missing limit" is $\sqrt{2} \notin \mathbb{Q}$.

Definition 2.7 (Complete Metric Space)**.** A metric space (X,d) is *complete* if every Cauchy sequence in X converges to a limit in X.

Theorem 2.5 (Completeness of $\mathbb{R}$)**.** *$(\mathbb{R}, |\cdot|)$ is a complete metric space.*

Proof. Let (x_n) be a Cauchy sequence in $\mathbb{R}$. First, (x_n) is bounded: there exists N such that $|x_m - x_n| < 1$ for all $m, n \geq N$, so $|x_n| \leq |x_N| + 1$ for $n \geq N$, giving $|x_n| \leq \max(|x_1|, \ldots, |x_{N-1}|, |x_N| + 1)$ for all n.

By the Bolzano–Weierstrass theorem (every bounded sequence in $\mathbb{R}$ has a convergent subsequence), there is a subsequence $x_{n_k} \to x$ for some $x \in \mathbb{R}$.

We show $x_n \to x$. Given $\varepsilon > 0$, choose N_1 such that $|x_m - x_n| < \varepsilon/2$ for $m, n \geq N_1$, and N_2 such that $|x_{n_k} - x| < \varepsilon/2$ for $k \geq N_2$. For $n \geq N_1$, pick k with $n_k \geq \max(N_1, n_{N_2})$. Then $|x_n - x| \leq |x_n - x_{n_k}| + |x_{n_k} - x| < \varepsilon/2 + \varepsilon/2 = \varepsilon$. □

Theorem 2.6 (Completeness of $\mathbb{R}^n$)**.** *$(\mathbb{R}^n, d_p)$ is complete for every $1 \leq p \leq \infty$.*

Proof. Let $(x^{(k)})_{k=1}^{\infty}$ be a Cauchy sequence in $(\mathbb{R}^n, d_p)$, where $x^{(k)} = (x_1^{(k)}, \ldots, x_n^{(k)})$. For each coordinate i, $|x_i^{(k)} - x_i^{(m)}| \leq \|x^{(k)} - x^{(m)}\|_p$ (since the ℓ^p norm dominates each coordinate), so $(x_i^{(k)})_{k=1}^{\infty}$ is Cauchy in $\mathbb{R}$. By Theorem 2.5, $x_i^{(k)} \to x_i$ for some $x_i \in \mathbb{R}$. Set $x = (x_1, \ldots, x_n)$. Then $\|x^{(k)} - x\|_p = \left(\sum_{i=1}^n |x_i^{(k)} - x_i|^p\right)^{1/p} \to 0$ as $k \to \infty$ (each term goes to 0; this is a finite sum). So $x^{(k)} \to x$. □

Definition 2.8 (Dense Subset, Separable Space)**.** A metric space (X, d) is *separable* if it contains a countable dense subset, i.e., there exists a countable set $D \subseteq X$ with $\mathrm{cl}(D) = X$.

Example 2.6. $\mathbb{R}^n$ is separable: $\mathbb{Q}^n$ is countable and dense. The space $C[a, b]$ with the sup metric is separable: by the Weierstrass approximation theorem, polynomials with rational coefficients form a countable dense subset.

Theorem 2.7 (Completion of a Metric Space)**.** *Every metric space (X, d) has a completion: there exists a complete metric space $(\hat{X}, \hat{d})$ and an isometry $\iota\colon X \to \hat{X}$ (i.e., $\hat{d}(\iota(x), \iota(y)) = d(x, y)$ for all $x, y \in X$) such that $\iota(X)$ is dense in $\hat{X}$. The completion is unique up to isometry.*

Proof. **Construction.** Let $\mathcal{C}$ be the set of all Cauchy sequences in X. Define an equivalence relation: $(x_n) \sim (y_n)$ if $d(x_n, y_n) \to 0$. Let $\hat{X} = \mathcal{C}/\!\sim$ be the set of equivalence classes, and write $[(x_n)]$ for the class of (x_n).

The metric is well defined. For $(x_n), (y_n) \in \mathcal{C}$, the sequence $(d(x_n, y_n))$ is Cauchy in $\mathbb{R}$ (since $|d(x_m, y_m) - d(x_n, y_n)| \leq d(x_m, x_n) + d(y_m, y_n)$ by the triangle inequality), hence convergent. Define $\hat{d}([(x_n)], [(y_n)]) = \lim_{n\to\infty} d(x_n, y_n)$. One checks that this is independent of representatives (if $(x_n) \sim (x_n')$ and $(y_n) \sim (y_n')$, then $|d(x_n, y_n) - d(x_n', y_n')| \leq d(x_n, x_n') + d(y_n, y_n') \to 0$) and satisfies the metric axioms.

Isometric embedding. Define $\iota(x) = [(x, x, x, \ldots)]$ (the class of the constant sequence). Then $\hat{d}(\iota(x), \iota(y)) = \lim_n d(x, y) = d(x, y)$, so ι is an isometry.

Density of $\iota(X)$. Given $[(x_n)] \in \hat{X}$ and $\varepsilon > 0$, choose N with $d(x_m, x_n) < \varepsilon/2$ for $m, n \geq N$. Then $\hat{d}([(x_n)], \iota(x_N)) = \lim_n d(x_n, x_N) \leq \varepsilon/2 < \varepsilon$.

Completeness of $\hat{X}$. Let (p_k) be Cauchy in $\hat{X}$. Since $\iota(X)$ is dense, for each k choose $z_k \in X$ with $\hat{d}(p_k, \iota(z_k)) < 1/k$. We claim (z_k) is Cauchy in X: indeed, $d(z_k, z_m) = \hat{d}(\iota(z_k), \iota(z_m)) \leq \hat{d}(\iota(z_k), p_k) + \hat{d}(p_k, p_m) + \hat{d}(p_m, \iota(z_m)) < 1/k + \hat{d}(p_k, p_m) + 1/m \to 0$. Let $p = [(z_k)] \in \hat{X}$. Then $\hat{d}(p_k, p) \leq \hat{d}(p_k, \iota(z_k)) + \hat{d}(\iota(z_k), p) < 1/k + \hat{d}(\iota(z_k), p) \to 0$ since $\hat{d}(\iota(z_k), p) = \lim_n d(z_k, z_n) \to 0$ as $k \to \infty$.

Uniqueness. If $(\hat{X}_1, \hat{d}_1, \iota_1)$ and $(\hat{X}_2, \hat{d}_2, \iota_2)$ are two completions, define $\varphi\colon \iota_1(X) \to \iota_2(X)$ by $\varphi(\iota_1(x)) = \iota_2(x)$. This is an isometry, hence uniformly continuous, and extends uniquely to an isometry $\hat{X}_1 \to \hat{X}_2$ by density and completeness. □

Completion and Function Approximation

In RL with function approximation, we often work with a parametric class of functions (e.g., neural networks) that is not complete under the sup norm. The completion theo-

rem guarantees that this class can be embedded into a complete space. Understanding which space we "complete to" is crucial for analyzing whether approximate value iteration converges.